MAT 1272

Sections 11.1–11.3
Chi-Square Tests

Chapter 11 Overview

Topics:

1. Chi-Square Distribution
2. Goodness-of-Fit Test
3. Test of Independence

Goal:

Use chi-square methods to analyze categorical data.

11.1 The Chi-Square Distribution

The chi-square distribution:

- Uses only positive values.
- Is skewed to the right.
- Depends on the degrees of freedom.
- Becomes more symmetric as degrees of freedom increase.

Notation:

$$\chi^2$$

(read "chi-square")

Degrees of Freedom

Degrees of freedom determine the shape of the distribution.
Examples:

- $df = 2$
- $df = 5$
- $df = 10$

As df increases, the distribution becomes less skewed.

When Do We Use Chi-Square Tests?

Chi-square tests are used for:

1. Goodness-of-Fit Tests
2. Tests of Independence

These tests work with categorical data.
Examples:

- Favorite color
- Political party
- Day of week
- Gender

Reminder: Using the p-value

The TI-84 will report a p-value.

Decision Rule:

$$\text{If } p \leq \alpha, \text{ reject } H_0$$

$$\text{If } p > \alpha, \text{ fail to reject } H_0$$

Remember:
A small p-value provides evidence against the null hypothesis.

11.2 Goodness-of-Fit Test

A goodness-of-fit test compares:

- Observed frequencies
- Expected frequencies

Question:
Do the observed results fit a claimed distribution?
Examples:

- Is a die fair?
- Are sales evenly distributed?
- Are survey responses consistent with previous results?

Observed and Expected Frequencies

Observed frequency:

$$O$$

Actual count collected from the sample.

Expected frequency:

$$E = np$$

where

- $n =$ sample size
- $p =$ hypothesized proportion

Chi-Square Test Statistic

The test statistic is

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

where

- O = observed frequency
- E = expected frequency

Large values of χ^2 suggest disagreement between observed and expected counts.

Degrees of Freedom for GOF

For a goodness-of-fit test:

$$df = k - 1$$

where

$$k = \text{number of categories}$$

Example:
A die has 6 categories.

$$df = 6 - 1 = 5$$

Conditions

To perform a goodness-of-fit test:

- Random sample
- Independent observations
- Expected frequency at least 5 in every category

Example: Fair Die

A die is rolled 60 times.
Observed frequencies:

$$7,\ 12,\ 8,\ 15,\ 11,\ 7$$

Question:
Do these results support the claim that the die is fair?

Steps for a Goodness-of-Fit Test

1. State hypotheses.
2. Find expected frequencies.
3. Compute χ^2.
4. Find the p-value.
5. Make a decision.

Decision rule:
Reject H_0 if

$$p < \alpha$$

TI-84: Goodness-of-Fit Test

Enter:

- Observed frequencies in L1
- Expected frequencies in L2

Then:
STAT
TESTS
Chi-Square GOF-Test
Fill in:

- Observed List = L1
- Expected List = L2
- df = k-1

Select:
Calculate

11.3 Contingency Tables

A contingency table summarizes two categorical variables.
Example:

	Own Phone	No Phone
Men	640	450
Women	440	470

Test of Independence

Question:
Are two categorical variables related?
Examples:

- Gender and phone ownership
- Gender and voting preference
- Age group and product preference

Hypotheses

Null hypothesis:

$$H_0 :$$

The variables are independent.

Alternative hypothesis:

$$H_1 :$$

The variables are dependent.

Expected Frequency Formula

For each cell:

$$E = \frac{\text{(Row Total)(Column Total)}}{\text{Grand Total}}$$

This formula is used to compute all expected counts.

Degrees of Freedom

For a test of independence:

$$df = (R - 1)(C - 1)$$

where

- R = number of rows
- C = number of columns

Example:
A 2×2 table:

$$df = (2 - 1)(2 - 1) = 1$$

Chi-Square Statistic

The same test statistic is used:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Large values indicate evidence of dependence.

TI-84: Test of Independence

Enter the contingency table into Matrix A.
Then:

- STAT
- TESTS
- Chi-Square Test

Choose:
Calculate
The calculator reports:

- χ^2
- p-value
- degrees of freedom

Decision Rule

Compare:

$$p$$

with

$$\alpha$$

If

$$p < \alpha$$

Reject H_0.
Otherwise:
Fail to reject H_0.

Final Exam Review

Know how to:

1. Identify the correct chi-square test.
2. Compute expected frequencies.
3. Find degrees of freedom.
4. Use the TI-84.
5. Interpret the p-value.

Questions?